THE LOWER SHRIEK ALONG THE PROJECTION TO THE POINT IS A LEFT INVERSE OF THE PULLBACK WHEN THE SITE HAS AN INITIAL OR A FINAL OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the evaluation of the lower shriek on an initial object, the evaluation of the pushforward on a final object giving Rpi_* pi^{-1} = id, and the formal conclusion through the criterion for full faithfulness of an adjoint.

Editorial note. The label lemma-initial-final names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source sites-cohomology.tex –label lemma-initial-final –start 10188 –end 10206 –local-id 3570 returned RESULT=OK (L3570 free, occupancy FREE, public interval clear). Candidate taken from operations/pool-cards/stacks-sites-cohomology-halfB.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Computation of the lower shriek and of the pushforward on a site with an initial or final object. Same chapter, the example computing the functors pi_! and Rpi_* for the projection to the point when the site has an initial or a final object. Supplies the evaluation of pi_! on an initial object and of Rpi_* on a final object used in the two cases.
- Full faithfulness of an adjoint whose unit or counit is an isomorphism. Categories, the lemma that a functor with a left adjoint whose counit is an isomorphism is fully faithful and the subsequent statement identifying the adjunction when both maps are isomorphisms. Supplies the conclusion that pi^{-1} is fully faithful when Rpi_* pi^{-1} = id, and then that the counit pi_! pi^{-1} -> id is an isomorphism.

Source reference. Original lemma 07RB: sources/stacks-chow/tags.txt, source lines 9861–9861. Prerequisite pointer invoked at proof line(s) 10203; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08PF: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 10016–10016. Prerequisite pointer invoked at proof

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

line(s) 10190; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Notation and assumptions as in Example 08PF. If $\mathcal{C}$ has either an initial or a final object, then $L\pi_! \circ \pi^{-1} = id$ on $D(Ab)$, resp. $D(B)$.*

Proof. If $\mathcal{C}$ has an initial object, then $\pi_!$ is computed by evaluating on this object and the statement is clear. If $\mathcal{C}$ has a final object, then $R\pi_*$ is computed by evaluating on this object, hence $R\pi_* \circ \pi^{-1} \cong \text{id}$ on $D(Ab)$, resp. $D(B)$. This implies that $\pi^{-1} : D(Ab) \to D(\mathcal{C})$, resp. $\pi^{-1} : D(B) \to D(\underline{B})$ is fully faithful, see Categories, Lemma 07RB. Then the same lemma implies that $L\pi_! \circ \pi^{-1} = \text{id}$ as desired. $\square$